

AI of the People, by the People, for the People: A Social Choice Approach to Collective Control of Artificial Intelligence

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Informatics



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Democratic control of AI

Existing work largely focuses on the **macro level**:
governance and institutional design (*regulatory control of AI*).

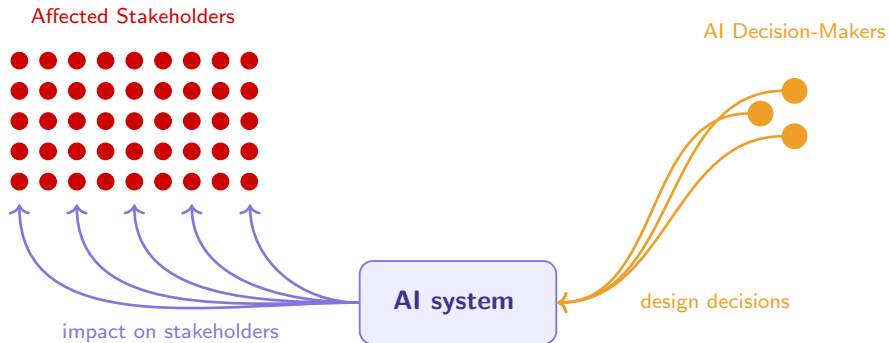
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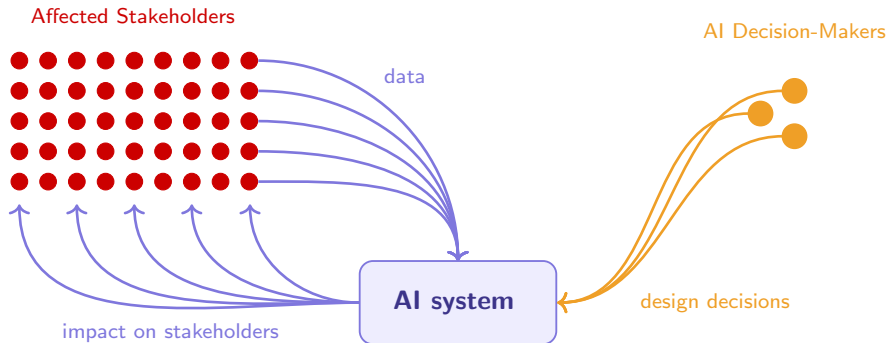
Our framework: collective control of AI
incorporating collective decision-making mechanisms directly into
the AI *development and deployment* process.

- ▶ the concrete technical and socio-technical **counterpart** to
regulatory control of AI

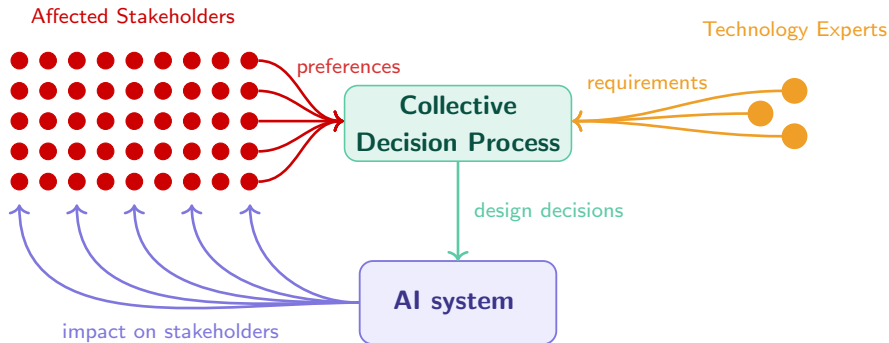
Status Quo



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Collective Decisions about AI



Goal: Collective Control of AI Systems

include affected stakeholders in AI development decisions

Background: Social Choice Theory

Social Choice Theory is the formal mathematical study of collective decision problems and preference aggregation rules.

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Central insight: Results of collective decisions heavily depend on the chosen preference aggregation rule.

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Axiomatic Method

Normative evaluation of preference aggregation rules

- ▶ proposition of normative axioms (e.g. anonymity)
- ▶ study of aggregation rules w.r.t. proposed axioms

1

What does it mean for an AI system to be aligned with collective preferences?

Framework for AI systems as social choice functions

Collective Control of AI – Two Perspectives

1

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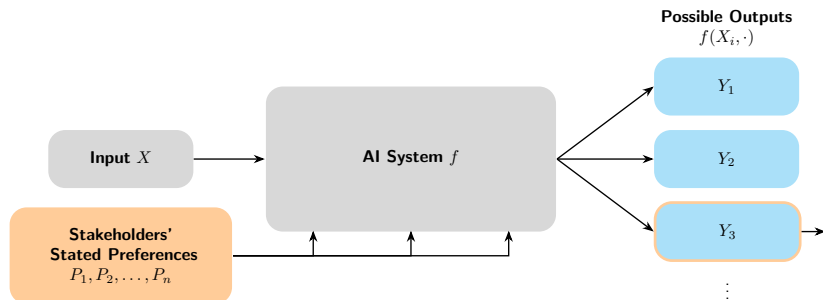
Framework for AI systems as social choice functions

2

How can we enable collective decision-making during AI development?

AI development decisions as concrete social choice problems

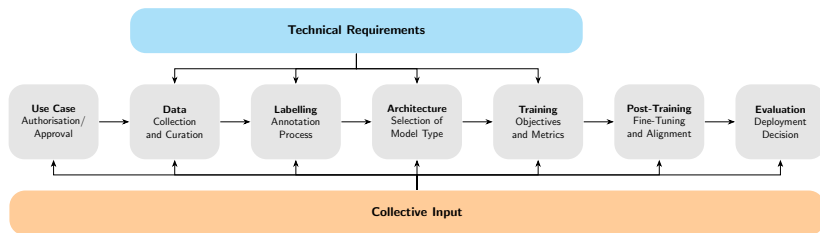
1 – AI Systems as Social Choice Functions



Social choice as lens for collective control of AI

- ▶ What *influence* can collective input have on AI outputs?
- ▶ How can stakeholders' preferences be *elicited* and *formalised*?
- ▶ Which *axioms* should AI systems satisfy w.r.t. the preferences?

2 – Collective Control in the ML Development Process



- ▶ How can we *enable collective decisions* in each stage?
- ▶ How can we *evaluate decisions* about the components?

We connect the decision problems in each stage to **established social choice problems**.

Open Questions and Ongoing Work

Stakeholder Identification

How can we identify the relevant stakeholders for a given system?

Preference Elicitation

How can laypeople meaningfully express their preferences about AI design?

⇒ COMSOC x social sciences



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Development of Concrete Decision-Making Mechanisms

Which concrete methods work well in the different steps? How can they be evaluated?

⇒ COMSOC x subsymbolic AI

